

1. Git Hands-On Lab Documentation

Experiment Name : Git Hands-On Lab (Git-HOL)

Step 1: Verify Git Installation

Command

git --version

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git version
git version 2.54.0.windows.1

deshm@Deshma MINGW64 ~/GitDemo (main)
$
```

Step 2: Configure Git Username and Email

Commands

```
git config --global user.name "Deshma U"
git config --global user.email "deshma2027@gmail.com"
```

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git config --global user.name "Deshma U"

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git config --global user.email "deshma2027@gmail.com"
```

Step 3: Verify Configuration

Command

git config --list

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
user.name=Deshma U
user.email=deshma2027@gmail.com
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
core.editor=notepad++.exe -multiInst -nosession
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

Step 4: Configure Notepad++ as Default Editor

Command

git config --global core.editor "notepad++"

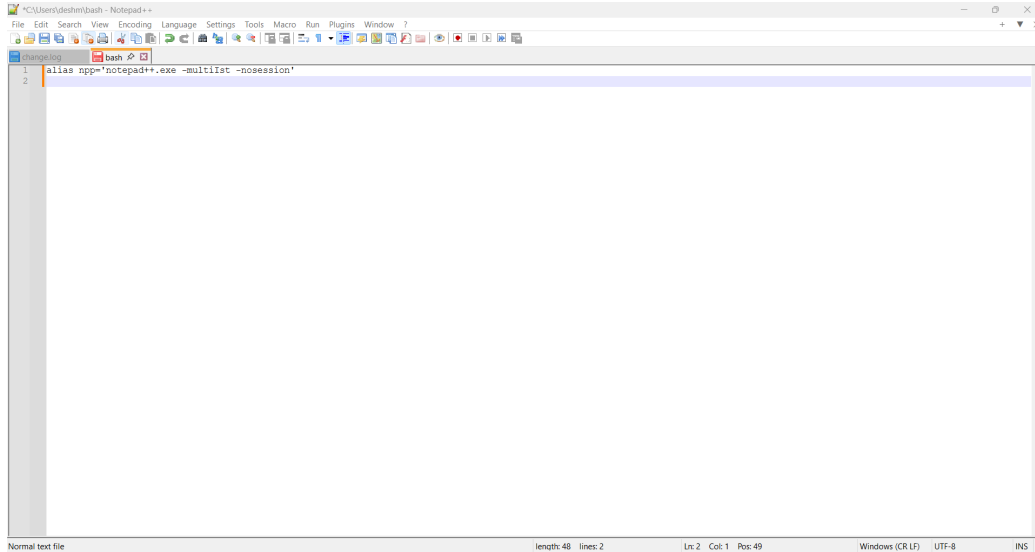
Output

```
deshm@Deshma MINGW64 ~
$ git config --global core.editor "notepad++.exe -multiInst -nosession"

deshm@Deshma MINGW64 ~
$ git config --global -e
```

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A screenshot of a Notepad++ window titled "C:\Users\deshm\bash - Notepad++". The window contains a single line of text: `alias npp='notepad++.exe -multitab -nosession'`. The status bar at the bottom indicates "Normal text file", "length: 48", "lines: 2", "Ln: 2", "Col: 1", "Pos: 49", "Windows (CR LF)", "UTF-8", and "INS".

Step 5: Create Git Repository

Commands

```
mkdir GitDemo
cd GitDemo
git init
```

Output

```
deshm@Deshma MINGW64 ~ (main)
$ cd GitDemo

deshm@Deshma MINGW64 ~/GitDemo (main)
$ pwd
/c/Users/deshm/GitDemo

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git init
Reinitialized existing Git repository in C:/Users/deshm/GitDemo/.git/
```

Step 6: Create welcome.txt File

Command

```
echo "Welcome to Git" > welcome.txt
```

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ echo "Welcome to Git" > welcome.txt
```

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Step 7: Check Git Status

Command

git status **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Step 8: Add File to Staging Area

Command

git add welcome.txt

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF the next time Git touches it

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   welcome.txt
```

Step 9: Commit File

Command

git commit -m "Added welcome.txt file"

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git commit -m "Added welcome.txt file"
>
> ^C

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git commit -m "Added welcome.txt file"
[main (root-commit) 69c13cc] Added welcome.txt file
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt
```

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Step 10: Push to Remote Repository

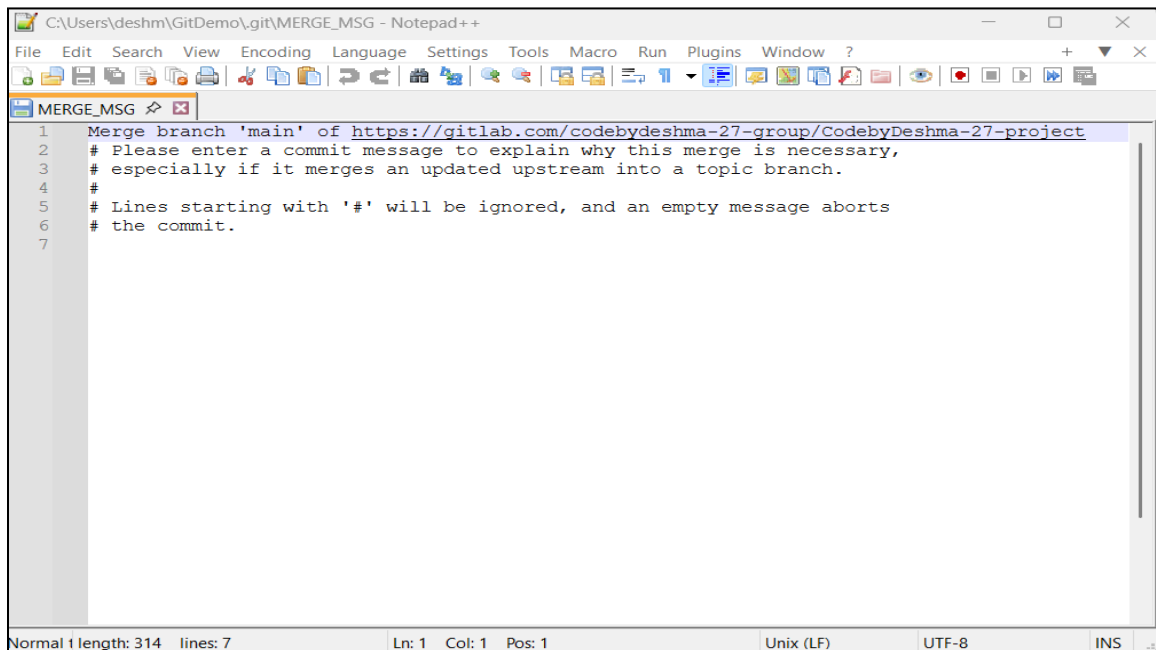
Commands

```
git remote add origin https://gitlab.com/CodebyDeshma-27-group/GitDemo.git
```

```
git pull origin main --allow-unrelated-histories
```

```
git push -u origin main
```

Notepad++

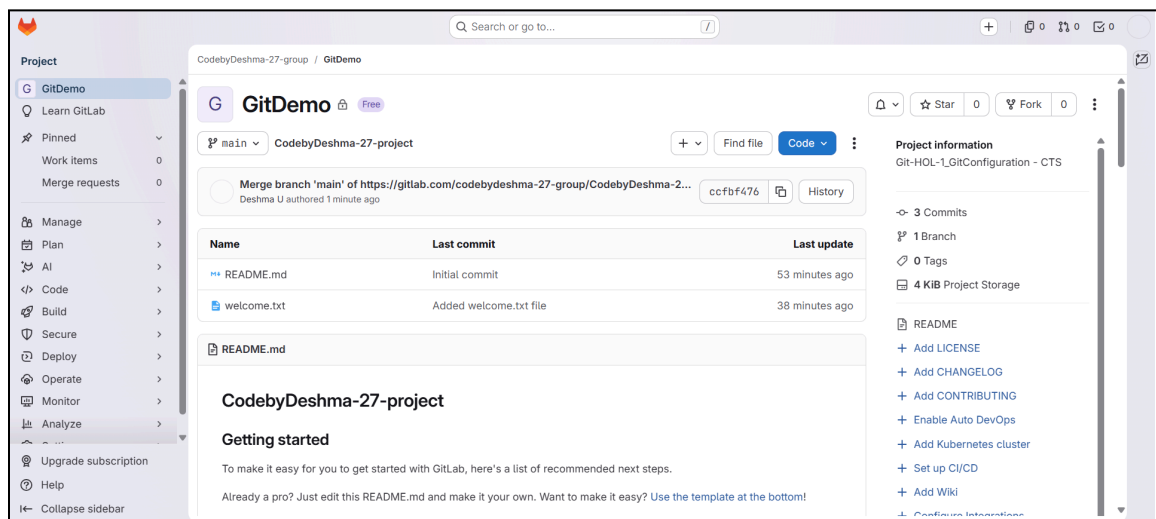


The screenshot shows a Notepad++ window titled "C:\Users\deshm\GitDemo\git\MERGE_MSG - Notepad++". The text inside is a merge message template:

```
1 Merge branch 'main' of https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project
2 # Please enter a commit message to explain why this merge is necessary,
3 # especially if it merges an updated upstream into a topic branch.
4 #
5 # Lines starting with '#' will be ignored, and an empty message aborts
6 # the commit.
7
```

The status bar at the bottom indicates: Normal 1 length: 314 lines: 7 Ln: 1 Col: 1 Pos: 1 Unix (LF) UTF-8 INS.

GitLab



The screenshot shows the GitLab web interface for the project "CodebyDeshma-27-group / GitDemo". The interface includes a sidebar with navigation options like Project, Pinned, Work items, Merge requests, Manage, Plan, AI, Code, Build, Secure, Deploy, Operate, Monitor, Analyze, Upgrade subscription, and Help. The main content area shows the project details, including the repository name, a merge button, and a table of files. The "Merge branch 'main'" button is highlighted. The table lists files: README.md (Initial commit, 53 minutes ago) and welcome.txt (Added welcome.txt file, 38 minutes ago). The README.md file is selected, showing its content. The right sidebar contains project information, including 3 commits, 1 branch, 0 tags, and 4 KIB Project Storage. It also lists links for README, Add LICENSE, Add CHANGELOG, Add CONTRIBUTING, Enable Auto DevOps, Add Kubernetes cluster, Set up CI/CD, Add Wiki, and Configure integrations.

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Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git remote add origin https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project.git

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git pull origin main --allow-unrelated-histories
info: please complete authentication in your browser...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 2.72 KiB | 174.00 KiB/s, done.
From https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project
* branch            main       -> FETCH_HEAD
* [new branch]       main       -> origin/main
Merge made by the 'ort' strategy.
 README.md | 93 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 93 insertions(+)
 create mode 100644 README.md

deshm@Deshma MINGW64 ~/GitDemo (main)
$ :wq
bash: :wq: command not found

deshm@Deshma MINGW64 ~/GitDemo (main)
$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 550 bytes | 550.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project.git
 474fdal..ccfbf47  main -> main
branch 'main' set up to track 'origin/main'.

deshm@Deshma MINGW64 ~/GitDemo (main)
$ |
```

Result Successfully performed Git configuration, repository creation, file tracking, commit, pull, and push operations using Git Bash.

<https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project>